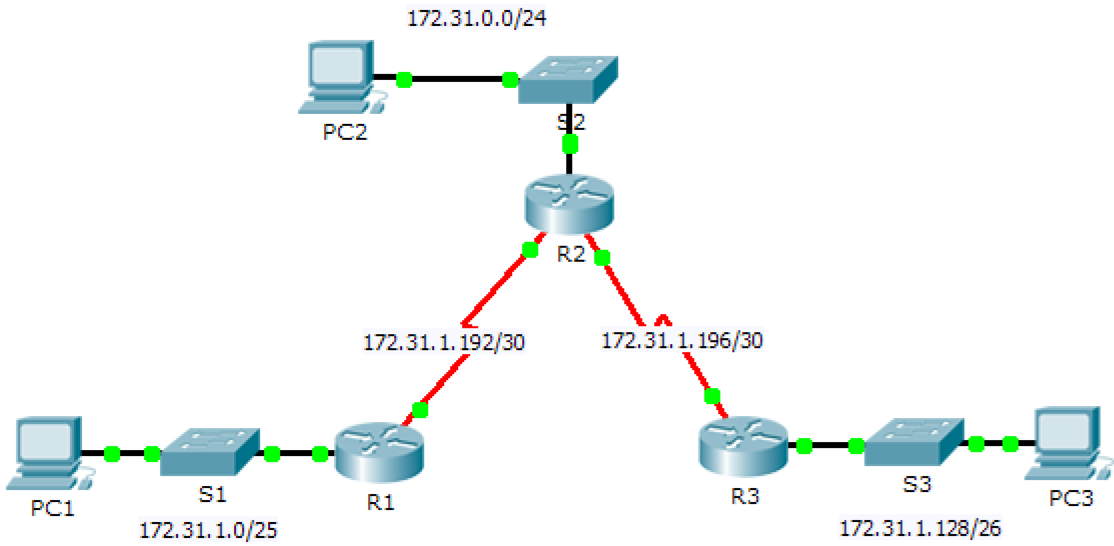
Packet Tracer - Configuring IPv4 Static and Default Routes

Topology



Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.31.1.1	255.255.255.128	N/A
	S0/0/0	172.31.1.194	255.255.255.252	N/A
R2	G0/0	172.31.0.1	255.255.255.0	N/A
	S0/0/0	172.31.1.193	255.255.255.252	N/A
	S0/0/1	172.31.1.197	255.255.255.252	N/A
R3	G0/0	172.31.1.129	255.255.255.192	N/A
	S0/0/1	172.31.1.198	255.255.255.252	N/A
PC1	NIC	172.31.1.126	255.255.255.128	172.31.1.1
PC2	NIC	172.31.0.254	255.255.255.0	172.31.0.1
PC3	NIC	172.31.1.190	255.255.255.192	172.31.1.129

Objectives

Part 1: Examine the Network and Evaluate the Need for Static Routing

Part 2: Configure Static and Default Routes

Part 3: Verify Connectivity

Background

In this activity, you will configure static and default routes. A static route is a route that is entered manually by the network administrator to create a reliable and safe route. There are four different static routes that are used in this activity: a recursive static route, a directly attached static route, a fully specified static route, and a default route.

Part 1: Examine the Network and Evaluate the Need for Static Routing

a. Looking at the topology diagram, how many networks are there in total? → 5 networks

b. How many networks are directly connected to R1, R2, and R3? → R1 (2 networks), R2 (3 networks), R3 (2 networks)

c. How many static routes are required by each router to reach networks that are not directly connected?
➔ R1 needs 3 static routes, R2 needs 2 static routes, R3 needs 3 static routes

d. Test connectivity to the R2 and R3 LANs by pinging PC2 and PC3 from PC1.

Why were you unsuccessful? → The pings were unsuccessful because the routers only have knowledge of directly connected networks, and no static or default routes have been configured to reach remote networks.

Part 2: Configure Static and Default Routes

Step 1: Configure recursive static routes on R1.

a. What is recursive static route?
→ A recursive static route is a route that specifies a next-hop IP address and requires the router to perform an additional lookup in the routing table to determine the outgoing interface.

b. Why does a recursive static route require two routing table lookups?

→ A recursive static route requires two routing table lookups because the router must first find the route to the destination network and then perform a second lookup to determine how to reach the specified next-hop IP address.

c. Configure a recursive static route to every network not directly connected to R1, including the WAN link between R2 and R3.

d. Test connectivity to the R2 LAN and ping the IP addresses of PC2 and PC3.

Why were you unsuccessful?

→ The pings were unsuccessful because only R1 was configured with static routes. R2 and R3 do not yet have routes to reach R1's network, so the return traffic cannot be delivered.

Step 2: Configure directly attached static routes on R2.

a. How does a directly attached static route differ from a recursive static route?

→ A directly attached static route specifies the exit interface and requires only one routing table lookup, whereas a recursive static route specifies a next-hop IP address and requires an additional lookup to determine the outgoing interface.

b. Configure a directly attached static route from R2 to every network not directly connected.

c. Which command only displays directly connected networks? → The command ‘show ip route connected’ displays only directly connected networks in the routing table.

d. Which command only displays the static routes listed in the routing table? → The command ‘show ip route static’ displays only the static routes in the routing table.

e. When viewing the entire routing table, how can you distinguish between a directly attached static route and a directly connected network?

   ➔ Directly connected networks are identified by the code “C”, while directly attached static routes are identified by the code “S” in the routing table.

Step 3: Configure a default route on R3.

a. How does a default route differ from a regular static route?
   ➔ A default route is used to forward packets to any destination network not found in the routing table, while a regular static route is used to route packets to a specific destination network.

b. Configure a default route on R3 so that every network not directly connected is reachable.

c. How is a static route displayed in the routing table?

   ➔ A static route is displayed with the code "S" in the routing table.

Step 4: Document the commands for fully specified routes.

Note: Packet Tracer does not currently support configuring fully specified static routes. Therefore, in this step, document the configuration for fully specified routes.

a. Explain a fully specified route.

   ➔ A fully specified route is a static route that includes both the exit interface and the next-hop IP address, allowing the router to forward packets without performing an additional lookup.

b. Which command provides a fully specified static route from R3 to the R2 LAN?
   ➔ The command is:
   ip route 172.31.0.0 255.255.255.0 s0/0/1 172.31.1.197
   which specifies both the exit interface and the next-hop IP address from R3 to reach the R2 LAN.

c. Write a fully specified route from R3 to the network between R2 and R1. Do not configure the route; just calculate it.
   ➔ The fully specified route from R3 to the network between R2 and R1 is:
   ip route 172.31.1.192 255.255.255.252 s0/0/1 172.31.1.197

d. Write a fully specified static route from R3 to the R1 LAN. Do not configure the route; just calculate it.
   ➔ The fully specified static route from R3 to the R1 LAN is:
   ip route 172.31.1.0 255.255.255.128 s0/0/1 172.31.1.197

Step 5: Verify static route configurations.

Use the appropriate **show** commands to verify correct configurations.

Which **show** commands can you use to verify that the static routes are configured correctly?

➔ The commands show ip route, show ip route static, and show running-config can be used to verify that static routes are configured correctly.

Part 3: Verify Connectivity

Every device should now be able to ping every other device. If not, review your static and default route configurations.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 1: Examine the Network and Evaluate the Need for Static Routing	a - d	10	
	Part 1 Total	**10**	
Part 2: Configure Static and Default Routes	Step 1	7	
	Step 2	7	
	Step 3	3	
	Step 4	10	
	Step 5	3	
	Part 2 Total	**30**	
	Packet Tracer Score	**60**	
	Total Score	**100**	